

# Inheritance and Extension: How CKS Builds on Knowledge Objects and OIDA Along the Multi-Human Axis

*A derivation note from the Coordination Knowledge Substrate (CKS) pattern.*

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**Date:** 27 April 2026

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## Attribution

This work derives from and formalizes the Coordination Knowledge Substrate (CKS) pattern introduced in *Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems* (Li, April 2026). It does not introduce new axioms. Its sole contribution is to consolidate, in a single derivation, the inheritance relationship the source paper establishes between CKS and two specific external prior-art lines — Knowledge Objects (KO) and OIDA — and to identify the precise axes on which CKS extends each. The central architectural claim it formalizes is that CKS’s relationship to OIDA is one of architectural difference along the multi-human axis, not addition.

## Abstract

The CKS pattern positions itself relative to two adjacent prior-art lines — Knowledge Objects (Zahn & Chana 2026) and OIDA (2026 preprint) — and the inheritance treatment is distributed across §5.2, §6.2, §8.2, and §9.4 of the source paper. This note consolidates that treatment. It states what CKS inherits from KO (the database-like cost model and the separation-of-concerns architecture between external storage and LLM processing) and how CKS extends KO on two axes (governance semantics; where disambiguation work lives), per §6.2. It states what CKS inherits from OIDA (signed-contradiction-edges as the basis for conflict-as-first-class substrate state) and how CKS differs from OIDA on three axes at the core-theory scope (governance as a design requirement; formal role and authority schema; AI-as-substrate-mediator), per §5.2. It then formalizes the central architectural claim from §9.4 — that CKS is **architecturally different from OIDA on the multi-human axis, not OIDA plus multi-human** — by stating precisely what that distinction means: the multi-human axis is the scope along which CKS’s already-distinct three-axis posture extends, not an axis on which CKS extends OIDA’s posture. The note ends with an operational test for whether a given system stands in the inheritance relationship the source paper claims as architecturally novel.

## 1. Why this inheritance needs to be formalized as its own note

The CKS pattern’s positioning against KO and OIDA is the most likely point of reviewer or implementor misreading, and the misreading has a specific shape. A reader who sees that CKS adopts external structured memory, signed contradiction edges, and human governance may conclude that CKS is “external structured memory plus governance” — a coherent description but a strictly weaker architectural claim than the one the source paper makes. A reader who sees that the source paper’s

extension scope (Claim 6) introduces multi-human coordination over substrate may conclude that CKS is “OIDA plus multi-human” — multi-human-access features added on top of OIDA’s existing posture. Both readings are addressed in the source paper, but the addresses are distributed across §5.2 (the OIDA differentiation at core scope), §6.2 (the KO inheritance with two-axis extension), §8.2 (the family-level positioning), and §9.4 (the extension-scope restatement of the OIDA claim).

This note’s job is to consolidate those treatments into a single derivation. It does not introduce new commitments; every claim is traceable to one of the four sections above. What it does is make the inheritance relationship legible as one continuous argument, so that downstream work that adopts, extends, composes with, or argues against CKS can address the inheritance directly rather than reconstructing it from the source paper’s distributed treatment. The architectural-novelty position CKS commits to depends on this inheritance being named precisely: a system that satisfies the architectural commitments only by accident — for example, by deploying a KO substrate under multi-user access controls — is not what the source paper claims is novel, and naming the inheritance relationship is what makes the difference visible.

## 2. What CKS inherits from KO, and how it extends KO

Knowledge Objects (Zahn & Chana 2026) is the closest empirical precedent for the CKS substrate. The source paper at §6.2 names the inheritance and the extension in precise terms.

**Inherited from KO.** Two commitments. First, the database-like cost model:  $O(1)$  retrieval via hash lookup, per-query token cost approximately constant from  $N=100$  through  $N=100,000$ , and a  $\sim 252\times$  per-query cost difference at  $N=7,000$  versus in-context memory that, in KO’s language, grows linearly as the corpus expands. Second, the separation-of-concerns architecture between external storage and LLM processing: facts live in addressable external storage where they cannot decay through compression or be lost at session boundaries, while the LLM retains query understanding, answer generation, and the extraction of new content from ongoing interaction. KO’s “context rot” vocabulary names the failure mode this separation prevents; CKS adopts that vocabulary as the named failure mode the source-of-truth commitment defends against.

**Not inherited from KO.** CKS does not adopt KO’s specific data models, type systems, or implementation choices. KO’s `provenance_metadata` field, its density-adaptive retrieval mechanism, and its `(subject, predicate)` structured-key disambiguation are KO design choices CKS treats as implementation territory it does not commit to. KO is also silent on whether the substrate is governed by humans in the authority sense; the architectural framing is about storage and processing, not about authority over substrate content.

**How CKS extends KO.** Per §6.2, two axes. The first axis is *governance semantics*: where KO commits to a single `provenance_metadata` field as its governance hook — adequate for attribution and audit at the retrieval-unit level — CKS encodes role and authority, approval states, decision rationale, and conflict state as first-class substrate content. The CKS substrate carries governance-relevant structure as schema, not as a single metadata field. The second axis is *where disambiguation work lives*: KO disambiguates computationally at retrieval time, via density-adaptive retrieval and structured-key

matching; CKS disambiguates through human-authored schema at the cell level, making the distinctions humans draw into substrate content encoded at authoring time rather than recovered by runtime algorithm. The two design stances reflect different philosophies about what the substrate is responsible for carrying. KO treats the substrate as storage over which algorithmic disambiguation operates; CKS treats the substrate as the authored artifact whose structure already carries the distinctions.

The two-axis extension is not “extra fields on top of KO.” The additional schema content exists because CKS commits to a different stance about where disambiguation and governance work sit — explicit in authored structure rather than implicit in retrieval algorithm or in a single attribution field.

### 3. What CKS inherits from OIDA, and how it differs at core scope

OIDA (2026 preprint) is the closest 2024–2026 architectural precedent for treating contradictions as first-class substrate state. The source paper at §5.2 names the inheritance and the differentiation in precise terms.

**Inherited from OIDA.** The architectural move of treating contradictions as substrate content rather than as error states. OIDA’s signed contradiction edges between conflicting Knowledge Objects are a structural realization of conflict-as-object with identity: the edge has its own schema presence, points to both conflicting objects, and persists in the substrate rather than existing as transient state consumed in a single inference pass. OIDA’s QUESTION-as-modeled-ignorance primitive is the analogous move on the missing-knowledge axis. CKS adopts the architectural pattern these primitives represent — substrate-level preservation of contradictions and missing knowledge as first-class addressable state — as the basis for Claim 3’s conflict-preservation commitment.

**Not inherited from OIDA.** CKS does not adopt OIDA’s specific Knowledge Object types, the importance-score-and-decay maintenance model, or the deterministic Knowledge Gravity Engine as its maintenance mechanism. These are OIDA design choices CKS treats as implementation territory it does not commit to. CKS also does not adopt OIDA’s framing of what the substrate is for: OIDA frames the substrate as an epistemically-structured object that an AI substrate client reasons over; CKS frames the substrate as a human-governed coordination artifact that AI mediates over.

**How CKS differs from OIDA at the core-theory scope.** Per §5.2, three architectural axes. The first axis is *governance*: CKS commits to human governance of the substrate as a design requirement — humans retain inspect, modify, and override rights over substrate content and orchestration rules at any time — whereas OIDA does not specify a governance architecture; it specifies a maintenance architecture that operates over the substrate deterministically but does not locate authority over substrate content in any particular actor. The second axis is *role and authority*: CKS encodes formal role and authority semantics at the cell level — who may decide what, on what evidence, under which policy — as substrate content the cell’s orchestration rules draw on; OIDA does not commit to a role/authority schema, and its Knowledge Objects carry epistemic class but not institutional provenance in the role-authority sense. The third axis is *the AI’s architectural role*: CKS frames AI as substrate mediator — helping humans read, write, interpret, and compose substrate content and executing cell

operations under human-authored orchestration rules — whereas OIDA frames AI as substrate client, consuming structured memory to answer queries.

These three axes are not gradations of a shared commitment; they are independent architectural moves that together describe a different design posture from the one OIDA takes. The source paper is explicit on this point: the adjacency between OIDA and CKS lies at the substrate-state level (contradictions-as-objects, modeled ignorance), not at the design-posture level (who governs, what schema, what role the LLM plays).

#### **4. The multi-human axis as the scope along which CKS’s distinct posture extends**

The architectural claim §9.4 protects is that CKS is **architecturally different from OIDA on the multi-human axis, not OIDA plus multi-human**. The inheritance and differentiation in §§2–3 above are the load-bearing context for that claim: CKS’s posture is already distinct from OIDA at the core-theory scope along three axes; the multi-human axis is the scope along which that *already-distinct* posture extends.

The “OIDA plus multi-human” reading would describe an architecture that takes OIDA as its base and adds multi-human-access as a feature on top — leaving OIDA’s governance posture, role/authority posture, and AI-as-substrate-client framing intact, and adding access controls so that more than one user can read and write. Under that reading, the multi-human axis is a deployment configuration: a way to operate OIDA in a team setting. The CKS reading is structurally different. The multi-human axis is the scope along which CKS’s three-axis-distinct posture extends — meaning that human governance, role/authority schema, and AI-as-substrate-mediator are already present at the single-human core, and the extension renders each of those commitments at the multi-human scope. Multi-human governance is not a feature added on top of an unchanged base; it is what each of the core-theory commitments already implies when more than one human holds authority over the same substrate.

The distinction matters because the CKS extension reshapes the core commitments rather than adding a parallel one. Conflict preservation gains a multi-human dimension: contradictions between humans operating under shared orchestration rules are themselves substrate content with attribution. Path retraceability gains a multi-human dimension: every authority exercise is attributable, and rule-authoring decisions are traceable to specific humans operating under shared or layered governance. AI-as-substrate-mediator gains a multi-human dimension: AI outputs materialize as substrate entries meant to be seen, challenged, and extended by humans in different roles across sessions, rather than as per-session assistance delivered to one user at a time. None of these renderings is a feature on top of an unchanged base; each is the multi-human-scope rendering of a commitment that already had architectural content at the single-human scope.

This is what “architecturally different on the multi-human axis, not OIDA plus multi-human” precisely commits to. The misreading the phrase preempts is the most plausible misreading at the scope where OIDA is silent and CKS is most distinctive, and the source paper protects the phrase across both §5.2 and §9.4 for that reason.

## 5. What the multi-human-axis extension requires of CKS substrates

The architectural difference §4 names is operationalized through specific properties at the substrate level. A substrate that satisfies the multi-human-axis extension supports the following.

*Authority distribution.* Multiple humans hold the inspect, modify, and override rights the human-governed commitment names. Each exercise of authority is attributed to the human who exercised it. Authority is not concentrated in a single privileged human as a precondition of substrate operation.

*Concurrent-write provenance.* When multiple humans write to the substrate, each write is attributed; conflicts between human writers are preserved as substrate-level contradictions per the conflict-preservation commitment, with provenance attached to the conflict itself. Resolution is not silently performed by the substrate or by an LLM operating over it.

*Resolution attribution.* When a human resolves a contradiction — whether by direct override, by editing substrate content, or by authoring an orchestration rule that the cell follows in subsequent executions — the resolution decision is itself substrate content, attributed to the human who made it, and reversible by other humans operating under the same governance.

*Rule authorship attribution.* Orchestration rules carry authorship. Rule changes are traceable to the humans who authored them. The path-retraceability commitment renders this not as a logging convention but as a substrate-level property: the rules governing cell behavior are inspectable substrate content with attribution.

*No single point of authority.* The substrate does not require a single privileged human as gatekeeper. Authority can be distributed across multiple humans operating under shared or layered orchestration rules, each within the scope the rules grant.

These five properties are derivations from the source paper, not additions to it. They are the multi-human-scope renderings of commitments the core theory already defends — Claim 1 (human governance), Claim 3 (conflict preservation), and Claim 6 (the multi-human extension of AI-as-substrate-mediator).

## 6. Operational test for the inheritance relationship

A system stands in the inheritance relationship CKS describes if and only if all of the following are true.

1. The substrate is external structured memory the LLM operates over, with the database-like cost model and the separation-of-concerns architecture between storage and processing (KO inheritance per §6.2).
2. The substrate encodes role and authority, approval states, decision rationale, and conflict state as first-class content, and the distinctions humans draw are carried in authored schema rather than recovered by retrieval-time algorithm (CKS extension over KO on the governance-semantics and disambiguation-locus axes, §6.2).

3. Contradictions in substrate content are first-class addressable objects with provenance attached to the contradiction itself (OIDA inheritance, per §5.2 and Claim 3).
4. The substrate is human-governed in the authority sense — inspect, modify, and override rights at any time over substrate content and orchestration rules — per the operational test in *Authority, Not Labor: A Precise Definition of “Human-Governed”* (Li 2026); authority over substrate content is not located in a deterministic maintenance mechanism or in any LLM operating over the substrate (CKS differentiation from OIDA at core scope, §5.2).
5. The substrate encodes formal role and authority semantics at the cell level, as substrate content the cell’s orchestration rules draw on (CKS differentiation from OIDA at core scope, §5.2).
6. The LLM operates as substrate mediator — helping humans read, write, interpret, and compose substrate content and executing cell operations under human-authored orchestration rules — rather than as a substrate client reasoning over a structured store (CKS differentiation from OIDA at core scope, §5.2).
7. Authority is distributed across multiple humans operating under shared or layered orchestration rules, with attribution for every authority exercise and resolution decision, per the five operational properties §5 names (CKS extension along the multi-human axis, §9.4).

A system that satisfies (1)–(3) but not (4)–(6) is in the broad inheritance lineage of KO and OIDA but does not instantiate the CKS architectural posture at the core-theory scope. A system that satisfies (1)–(6) but not (7) is single-human CKS — coherent and within the source paper’s core-theory scope, but not in the extension-scope architectural relationship §9.4 names. A system that satisfies (1)–(7) stands in the inheritance relationship the source paper claims as architecturally novel.

## 7. Scope of the claim, and why naming this inheritance precisely matters

The inheritance relationship this note formalizes is bounded. It does not claim that CKS is the only architecture that extends KO or OIDA in interesting ways; both lines admit other extensions along other axes, and the source paper’s positioning against them is not a claim of exclusive succession. It does not claim that CKS subsumes KO or OIDA; both lines remain coherent on their own terms for their own use cases, and CKS draws on each for the specific commitments §§2–3 above name. It does not claim that KO or OIDA are inadequate as architectures; CKS extends and differs from them for a specific architectural goal — multi-human-governed coordination at the coordination and decision layer — not as a general improvement. It does not claim that KO or OIDA cannot be operated under multi-human governance; the source paper’s claim is about architectural commitments, not deployment options. A KO or OIDA substrate operated by multiple users with appropriate access controls is a coherent system; it is not a CKS substrate, because the architecture does not commit to the multi-human-axis extension as part of what makes the substrate what it is. The note also does not extend KO or OIDA on axes the source paper does not treat. Both lines are external prior art the source paper inherits from on specific axes; they are not primary sources for this note, and the note respects their authorship boundaries.

Within those bounds, naming this inheritance precisely is what makes the CKS architectural-novelty position defensible. Implementations that read CKS as “external structured memory with governance”

miss the multi-human axis and produce single-user systems that satisfy the architectural commitments only by accident. Implementations that read CKS as “OIDA with multi-user access” miss the architectural-difference claim and produce systems that bolt multi-user features onto OIDA without reorganizing the commitments. CKS is neither: it inherits from KO on storage and cost-model commitments and extends KO on governance semantics and disambiguation locus; it inherits from OIDA on the architectural treatment of contradictions and differs from OIDA on three axes at the core-theory scope; and the multi-human axis is the scope along which the resulting three-axis-distinct posture extends, rendering each core commitment at the multi-human scope rather than adding a multi-human feature to an unchanged base. The phrase **architecturally different from OIDA on the multi-human axis, not OIDA plus multi-human** names that distinction in the form most resistant to misreading, and the inheritance treatment in §§2–3 is what makes it defensible: at the scope where OIDA is silent, CKS is not adding multi-human features to OIDA but extending a substrate that was already structurally different from OIDA at the core-theory scope into multi-human coordination.

Subsequent work that adopts the CKS pattern, extends it, composes it with adjacent patterns, or argues against it should address the inheritance relationship as formalized here. Subsequent work that uses the inheritance relationship differently is using a different concept, and the difference should be named.

### Source paper

Li, W. (2026). *Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems*. April 2026. ORCID: 0009-0004-8065-3235.

### Companion note

Li, W. (2026). *Authority, Not Labor: A Precise Definition of “Human-Governed” in the Coordination Knowledge Substrate Pattern*. 24 April 2026. ORCID: 0009-0004-8065-3235.

### How to cite this note

Li, W. (2026). *Inheritance and Extension: How CKS Builds on Knowledge Objects and OIDA Along the Multi-Human Axis*. 27 April 2026. ORCID: 0009-0004-8065-3235.